Hints: Methods of proof – 4. Invariants, monovariants and double counting

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Treat the empty cell as a sixteenth tile, so a position is a permutation π of the 16 cells. Every move is a transposition of the empty cell with a neighbouring tile. What does a move do to $\operatorname{sgn}\pi$, and what does it do to the colour of the empty cell in the chessboard colouring?

Hint 2. Each move changes both: the sign flips and the empty cell changes colour. So $I = \operatorname{sgn}\pi \cdot (-1)^c$, where c is the colour (equivalently, the parity of the taxicab distance of the empty cell from its home), is invariant.

Hint 3. Swapping two tiles with the empty cell at home changes $\operatorname{sgn}\pi$ but not c, so it changes I: unreachable. General principle: the reachable configurations from x form the orbit of x under the group (or semigroup) generated by the moves; any function constant on these orbits, typically $x \mapsto \chi(g)$ for a homomorphism χ (here to $\{\pm 1\}$) that is trivial on all moves, separates orbits.

Problem 2

Note. "Decide which multisets are reachable" has no clean complete answer for three or more numbers (the group generated by the moves is infinite); the hints give the invariant, the full answer for two numbers, and what can be said about the second operation.

Hint 1. The map $(a, b) \mapsto (\frac{a+b}{\sqrt{2}}, \frac{a-b}{\sqrt{2}})$ is given by an orthogonal 2×2 matrix. Which simple symmetric functional of the numbers on the board does it preserve?

Hint 2. The sum of squares is invariant, so a multiset with a different sum of squares is unreachable. It is not sufficient: the move is an involution, so with two numbers the only multisets reachable from $\{a, b\}$ are $\{a, b\}$ and $\{\frac{a+b}{\sqrt{2}}, \frac{a-b}{\sqrt{2}}\}$; in general only countably many multisets are reachable while the sphere $\sum x_i^2 = \text{const}$ is uncountable.

Hint 3. For $(a, b) \mapsto (a + b, ab)$: the new pair is the root pair of $(x - s)(x - p)$, where $s = a + b$, $p = ab$, so on (s, p) the operation acts as $(s, p) \mapsto (s+p, sp)$; no symmetric polynomial is preserved. Use a monovariant instead: the sum grows by ab, so on positive numbers the sum strictly increases, no configuration recurs, and only multisets with larger sum (and positive entries) can be reached; a 0 on the board is frozen, since $(a, 0) \mapsto (a, 0)$.

Problem 3

Hint 1. For positive integers $a \geqslant b > 0$ the sum $a + b$ (or the larger number) strictly decreases at each step and is a positive integer, so the process stops, namely when one number becomes 0.

Hint 2. The invariant is $\gcd(a, b) = \gcd(a - b, b)$; at the end it is the remaining non-zero number. Grouping consecutive subtractions of the same b gives division with remainder: the Euclidean algorithm.

Hint 3. Complexity: subtraction alone can take a steps (e.g. $(a, 1)$). With division, if $r_k = qr_{k+1} + r_{k+2}$, then $r_k \geqslant r_{k+1} + r_{k+2}$, so the remainders dominate Fibonacci numbers backwards; the number of division steps is $O(\log b)$ (Lamé: at most about $\log_\varphi b + 1$; also $r_{k+2} < r_k/2$).

Hint 4. One continuous analogue: for the gradient flow $x' = -\nabla V(x)$ the energy V is a Lyapunov function, $V(x(t)) = V(x(0)) - \int_0^t |x'(s)|^2\,ds$. If V is bounded below, $\int_0^\infty |x'|^2 < \infty$, which is the continuous form of "a bounded monovariant cannot decrease forever". (Note also that Euclid on reals a, b terminates iff $a/b \in \mathbb{Q}$.)

Problem 4

Hint 1. $\sum_v \deg v = 2|E|$: count pairs (v, e) with v an endpoint of e. For the 4-cycle bound count the "cherries" $(v, \{x, y\})$ with $x \neq y$ both adjacent to v.

Hint 2. By vertices v there are $\sum_v \binom{d_v}{2}$ cherries; by pairs $\{x, y\}$ there are at most $\binom{n}{2}$, because without a C_4 two vertices have at most one common neighbour. Hence $\sum_v \binom{d_v}{2} \leqslant \binom{n}{2}$.

Hint 3. By convexity (Jensen), $n\binom{d}{2} \leqslant \binom{n}{2}$ with $d = 2|E|/n$, i.e. $d^2 - d - (n-1) \leqslant 0$, so $d \leqslant \frac{1+\sqrt{4n-3}}{2}$ and $|E| = \frac{nd}{2} \leqslant \frac{1}{2}\big(1 + \sqrt{4n-3}\big) \cdot \frac{n}{2}$.

Hint 4. Oddtown: sets $A_1, \ldots, A_m \subset \{1, \ldots, n\}$ with odd sizes and even pairwise intersections. Their characteristic vectors over $\mathbb{F}_2$ have Gram matrix I_m, which has rank m, so the vectors are independent and $m \leqslant n$. Comparison: double counting bounds a quantity by averaging over incidences (and the C_4 bound is asymptotically sharp); for Oddtown a counting argument gives only weaker bounds, while the rank gives $m \leqslant n$ at once, which is sharp (take singletons).

Problem 5

Hint 1. Encode S by a generating function: $|S|$ is the coefficient of x^{3860} in $f(x)^{2018}$, where $f(x) = x + x^2 + x^3 + x^4 + x^5 + x^6 + x^{10}$.

Hint 2. All coefficients are non-negative, so for every $x > 0$ we have $|S|\, x^{3860} \leqslant f(x)^{2018}$, i.e. $|S| \leqslant f(x)^{2018}/x^{3860}$. Choose x cleverly.

Hint 3. Take $x = \frac{1}{2}$: $f(\frac{1}{2}) = \frac{63}{64} + \frac{1}{1024} = \frac{1009}{1024} = \frac{2018}{2048}$, which gives exactly $|S| \leqslant 2^{3860}\big(\frac{2018}{2048}\big)^{2018}$.